DESCRIPTIVE STATISTICS · ILLUSTRATED LECTURE SLIDES

Measures of Variability: Variance

11 lecture slides, explained step by step and checked against the spreadsheet

PART A · Slides 1–4

Sample vs. population

Why two kinds of data need two formulas

PART B · Slides 5–7

Mean and variance

What variance measures and its formulas

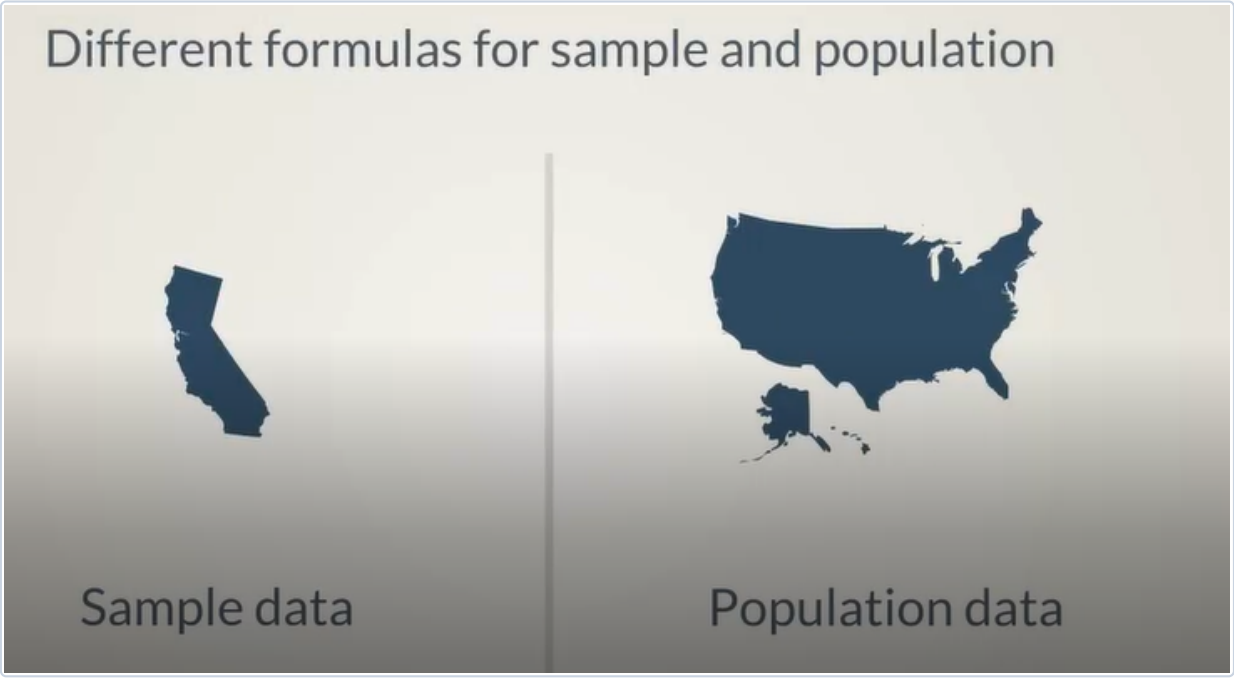
PART C · Slides 8–11

Worked example

Computing variance for 1, 2, 3, 4, 5

Page 13 reproduces the Excel workbook · Page 14 is a one-page cheat sheet

Two kinds of data, two formulas



ORIGINAL SLIDE 1

PICTURE IT

In the lecture the whole United States stands for the **population**, and a single state such as California stands for a **sample** taken from it.

IN PLAIN WORDS

Before you calculate anything, ask one question: **do I have every member of the group, or only part of it?** The answer decides which formula you use.

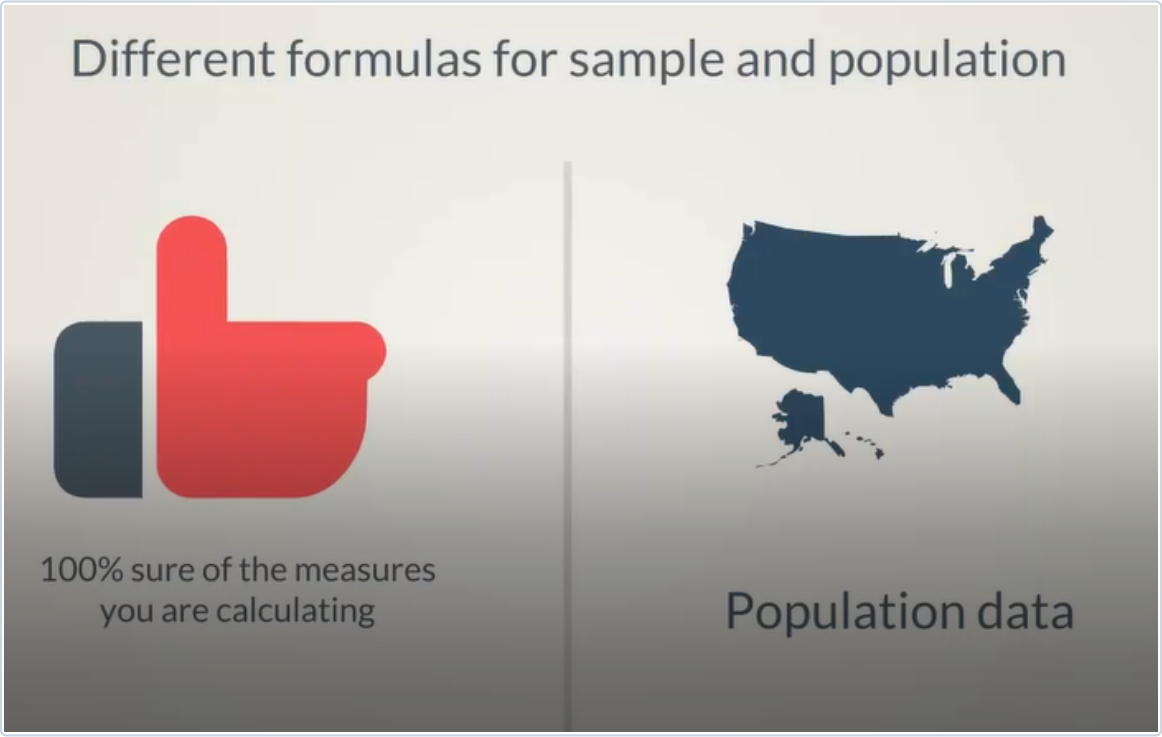
KEY IDEA

Decide first: sample or population? The formulas differ.

	Sample	Population
Covers	A part of the group	The whole group
Size	n	N
Mean	$\bar{x}$	μ
Variance	s^2	σ^2

NEXT › Population data: no guesswork

Population data: no guesswork



ORIGINAL SLIDE 2

VOCABULARY

A measure that describes a whole population is called a **parameter**. It is written with Greek letters: μ for the mean and σ^2 for the variance.

IN PLAIN WORDS

When your data **is** the whole population, every number you compute is exact. Nothing is estimated, because nothing is missing.

KEY IDEA

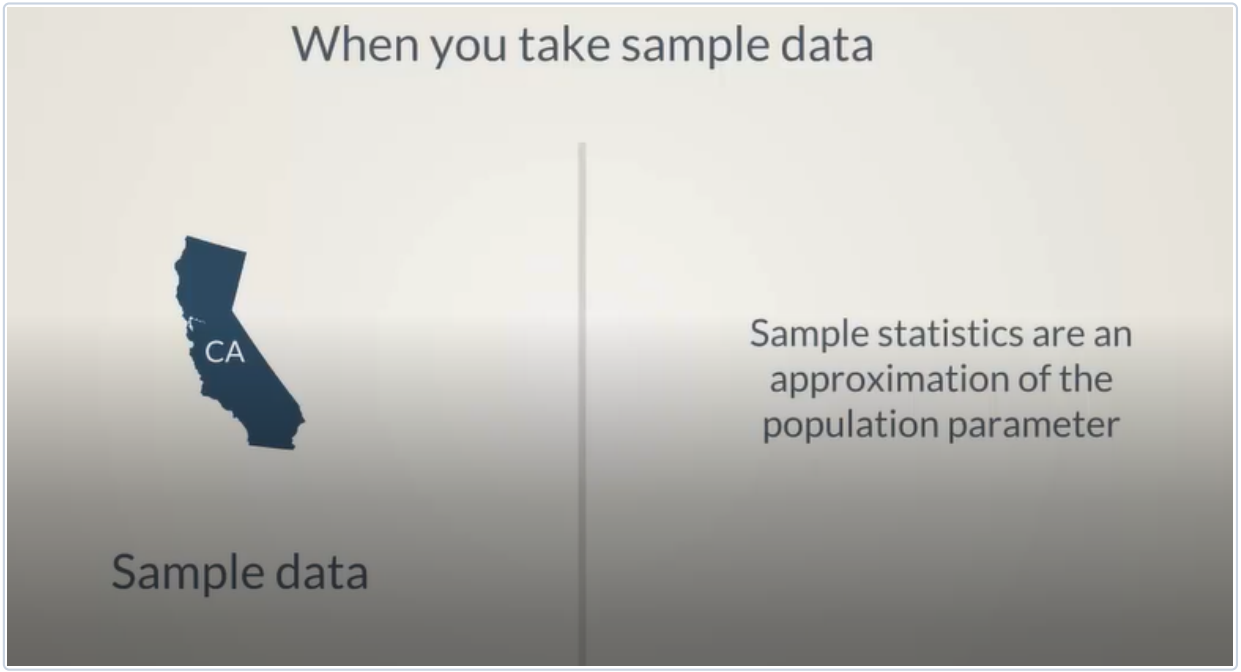
Population measures are exact parameters: you are 100% sure of them.

LINK TO THE SPREADSHEET

The workbook treats **1, 2, 3, 4, 5** as a complete population, so its results (mean 3, variance 2) are exact.

NEXT › Sample data: an approximation

Sample data: an approximation



ORIGINAL SLIDE 3

WHY IT MATTERS

California is just one of many possible samples of the United States. Any other state, or any mix of states, would give a slightly different answer.

IN PLAIN WORDS

Most of the time you cannot measure everyone, so you measure a **sample** and use it to learn about the population. The result is a good guess, but still a guess.

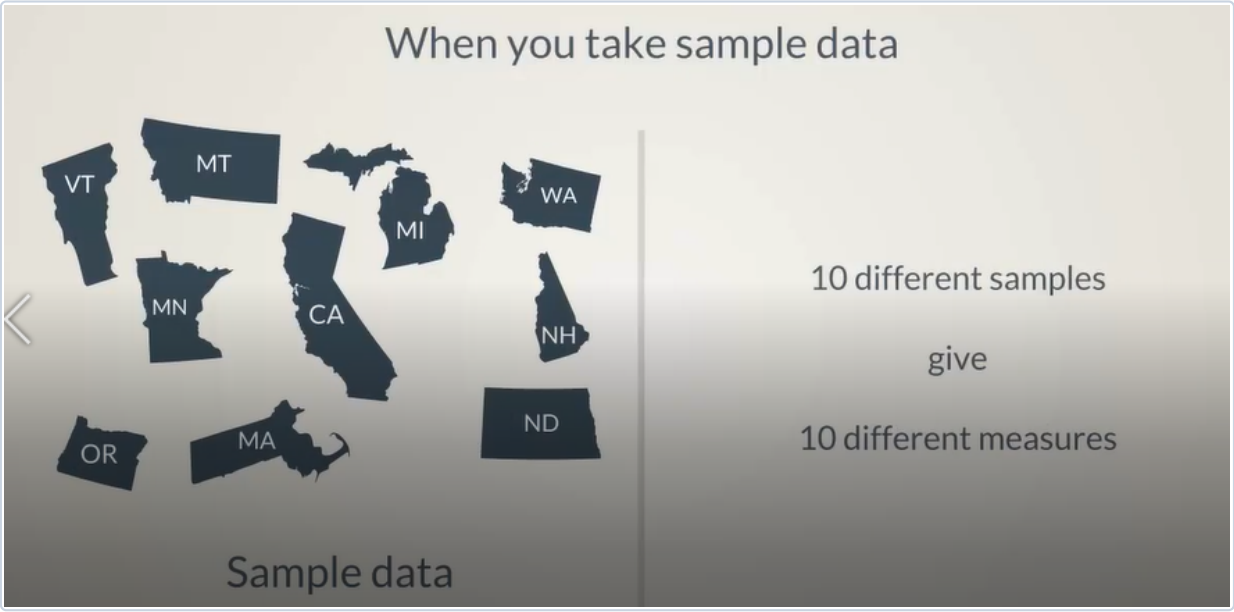
KEY IDEA

A sample statistic approximates the population parameter. It is not equal to it.

	Parameter	Statistic
Comes from	Whole population	A sample
Exact?	Yes	No, an estimate
Symbols	μ, σ^2	$\bar{x}$, s^2

NEXT › Different samples, different results

Different samples, different results



ORIGINAL SLIDE 4

TRY IT: ALL 10 SAMPLES OF SIZE 3 FROM THE POPULATION 1, 2, 3, 4, 5

Sample	$\bar{x}$	s^2	Sample	$\bar{x}$	s^2
(1,2,3)	2.00	1.00	(1,4,5)	3.33	4.33
(1,2,4)	2.33	2.33	(2,3,4)	3.00	1.00
(1,2,5)	2.67	4.33	(2,3,5)	3.33	2.33
(1,3,4)	2.67	2.33	(2,4,5)	3.67	2.33
(1,3,5)	3.00	4.00	(3,4,5)	4.00	1.00

The sample means range from **2.00 to 4.00** and the sample variances from **1.00 to 4.33**. The true population values are $\mu = 3$ and $\sigma^2 = 2$, so no single sample hits them exactly.

IN PLAIN WORDS

Draw ten different samples and you get ten different answers. This is called **sampling variation**, and it is why a sample statistic can only approximate the population.

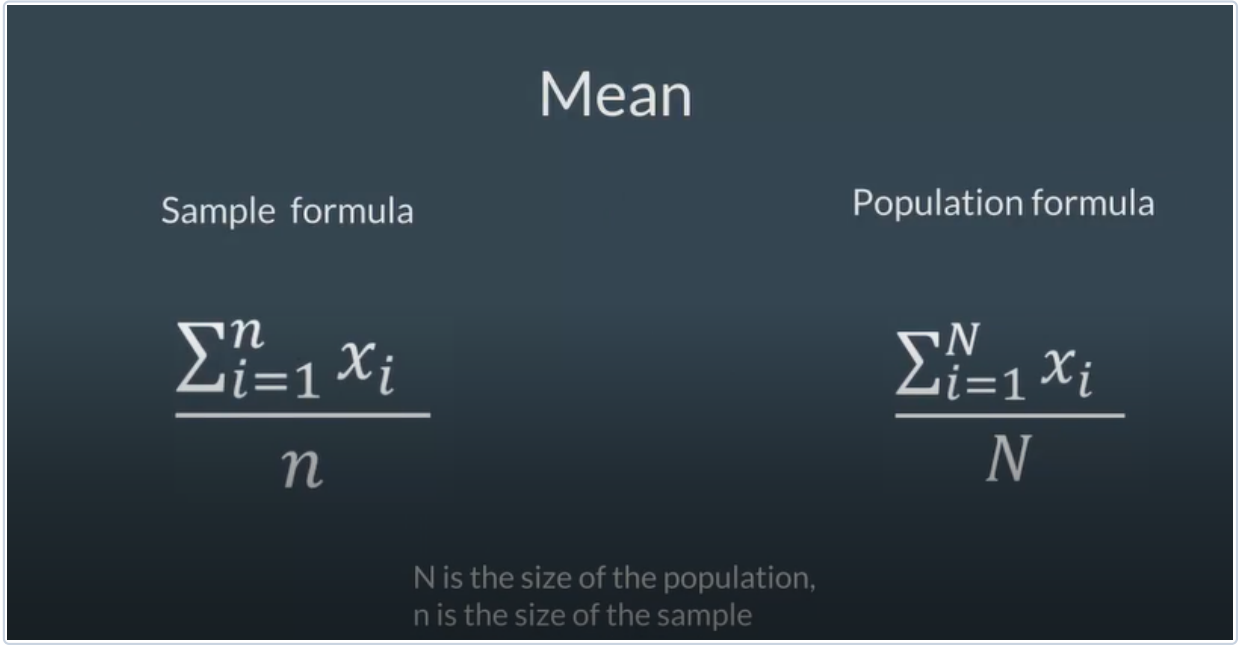
KEY IDEA

10 different samples give 10 different measures.

ADDED FOR ILLUSTRATION

This table is not from the lecture. I generated it from the spreadsheet's population to demonstrate the point.

NEXT › The mean: where we measure from



ORIGINAL SLIDE 5

NOTATION

N is the size of the population and **n** is the size of the sample, as the slide notes at the bottom. Capital for the whole group, lowercase for the part.

IN PLAIN WORDS

Variance measures spread **around the mean**, so the mean always comes first. The two formulas look different but work the same way: **add everything up, then divide by how many values there are.**

KEY IDEA

Sample: divide by n. Population: divide by N.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

sample mean

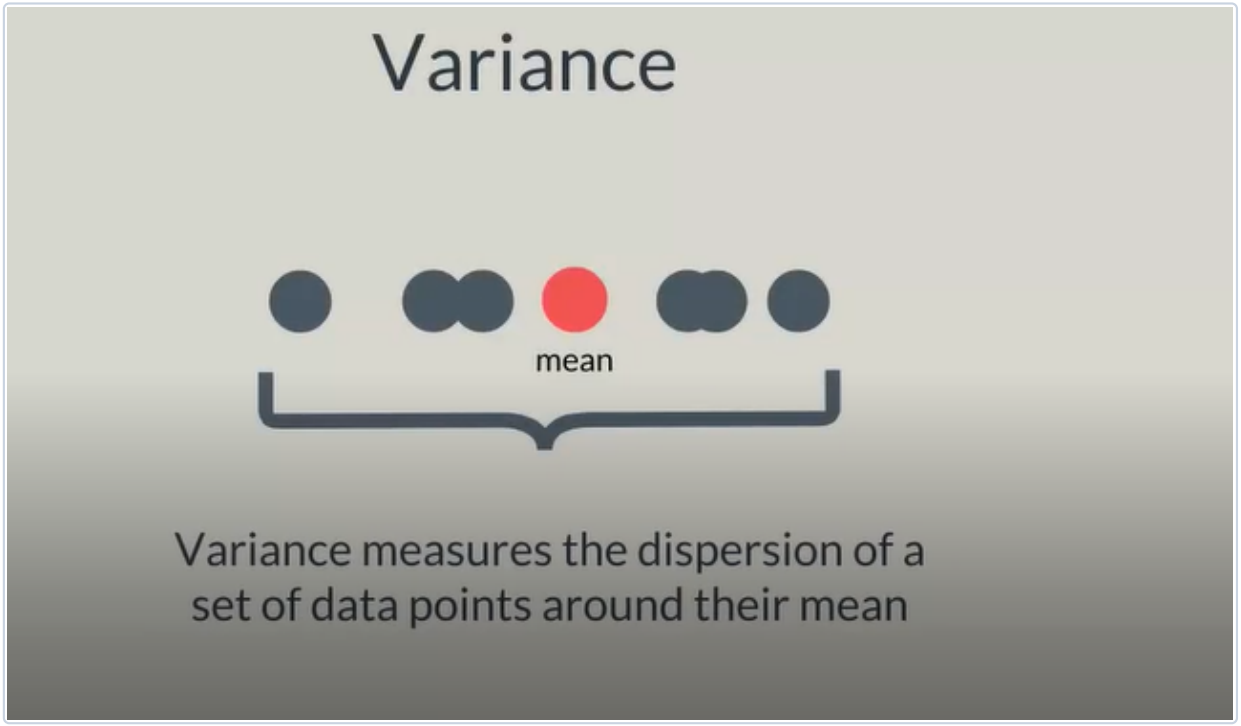
$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

population mean

FROM THE SPREADSHEET

`=AVERAGE(B4:B8)` on the values 1 to 5 gives **3**.

NEXT › Variance: the idea



ORIGINAL SLIDE 6

WHAT GOES INTO IT

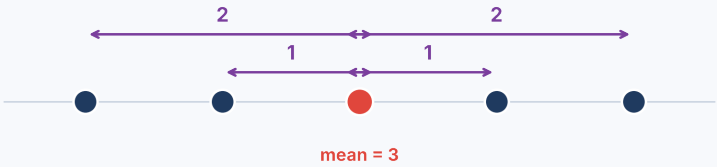
Variance is built from these distances, called **deviations**. Here they are 2, 1, 0, 1, 2 (distances are positive). The next slides turn them into one number.

IN PLAIN WORDS

Variance puts a number on how scattered the data points are around their mean. Tightly packed values give a small variance. Widely scattered values give a large one.

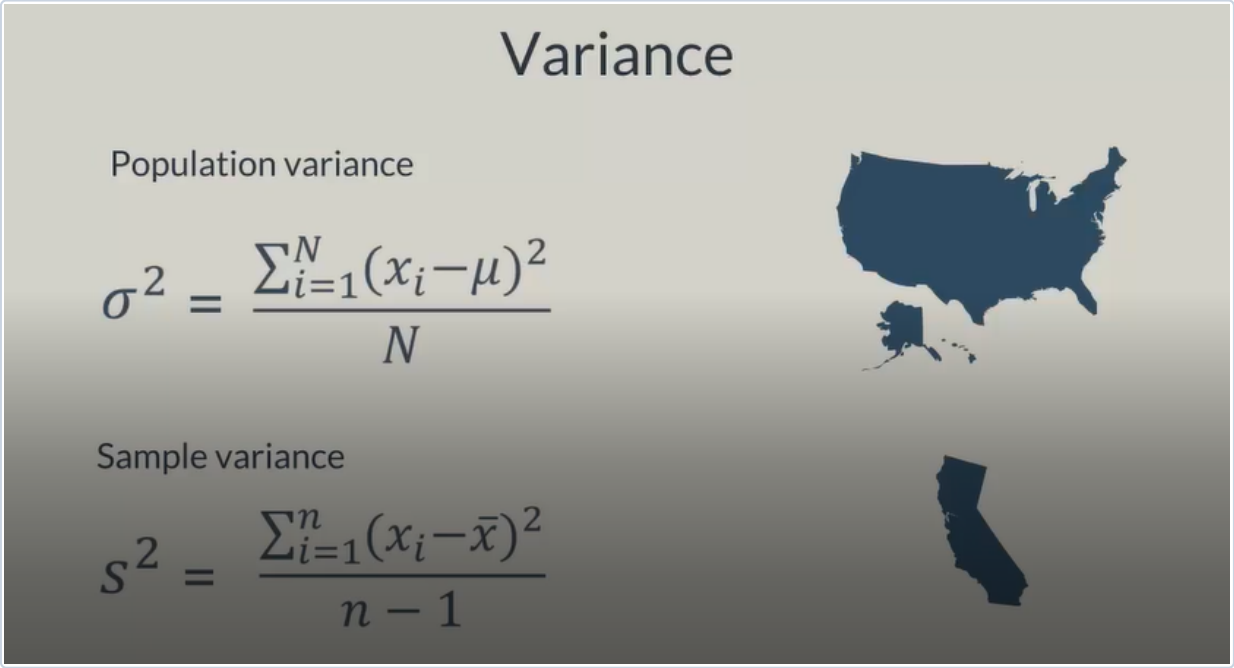
KEY IDEA

Variance measures the dispersion of a set of data points around their mean.



Distance of each of 1, 2, 4, 5 from the mean 3

NEXT › Variance: the two formulas



ORIGINAL SLIDE 7

WHY THE SQUARES

Why square? Distance cannot be negative, and squaring makes every term positive. Without it, values below the mean would cancel values above it, and the total would always be 0.

IN PLAIN WORDS

Take each value's distance from the mean, **square** it, add the squares, then divide. The only difference between the formulas is the denominator.

KEY IDEA

Population: divide by N. Sample: divide by n − 1.

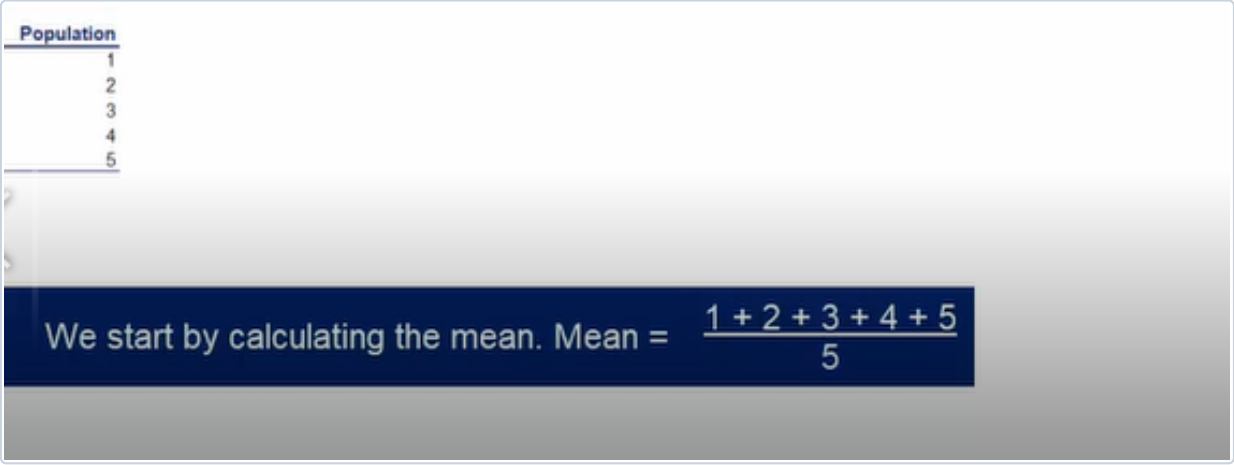
population

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i-\mu)^2}{N}$$

sample

$$s^2 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})^2}{n-1}$$

NEXT › Step 1: find the mean



ORIGINAL SLIDE 8

READING THE SLIDE

The slide's spreadsheet shows the **Population** column (1 to 5) and the formula banner. Next the mean (3.00) will appear beside it.

IN PLAIN WORDS

Our data is **1, 2, 3, 4, 5**. Every variance calculation starts the same way: work out the mean.

KEY IDEA

Mean = (1 + 2 + 3 + 4 + 5) ÷ 5 = 3

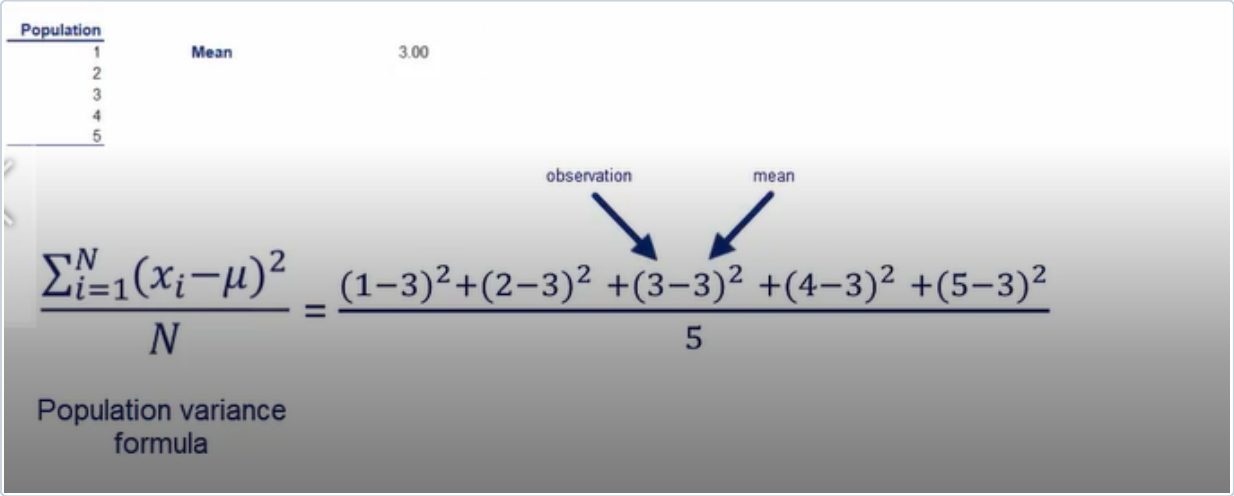
$$\mu = \frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

SPREADSHEET: AFTER THIS STEP

B	D	E
1	Mean	3
2		
3		
4		
5		

NEXT › Step 2: plug the mean into the formula

Step 2: plug the mean into the formula



ORIGINAL SLIDE 9

NUMERATOR, TERM BY TERM

Numerator: $(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2 = 4 + 1 + 0 + 1 + 4 = \mathbf{10}$.

IN PLAIN WORDS

Substitute each observation and the mean into the numerator. The arrows on the slide point to the observation and the mean in the third term, $(3 - 3)^2$.

KEY IDEA

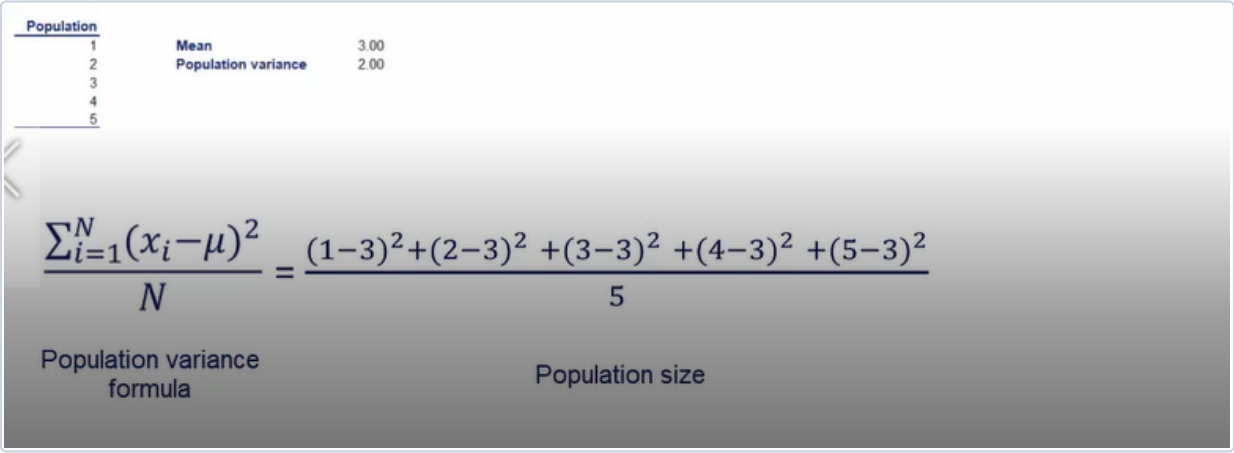
Subtract the mean from each value, then square.

x	x − 3	(x − 3)²
1	−2	4
2	−1	1
3	0	0
4	1	1
5	2	4
Sum	**0**	**10**

The raw differences add to **0**, so squaring is what makes a useful total of **10**.

NEXT › Step 3: divide by N to get σ²

Step 3: divide by N to get σ²



ORIGINAL SLIDE 10

CHECK

The slide's spreadsheet now shows **Mean 3.00** and **Population variance 2.00**, matching our hand calculation.

IN PLAIN WORDS

For the **population** formula, divide the numerator (10) by the population size **N = 5**. The slide labels this denominator "Population size".

KEY IDEA

Population variance σ² = 10 ÷ 5 = 2

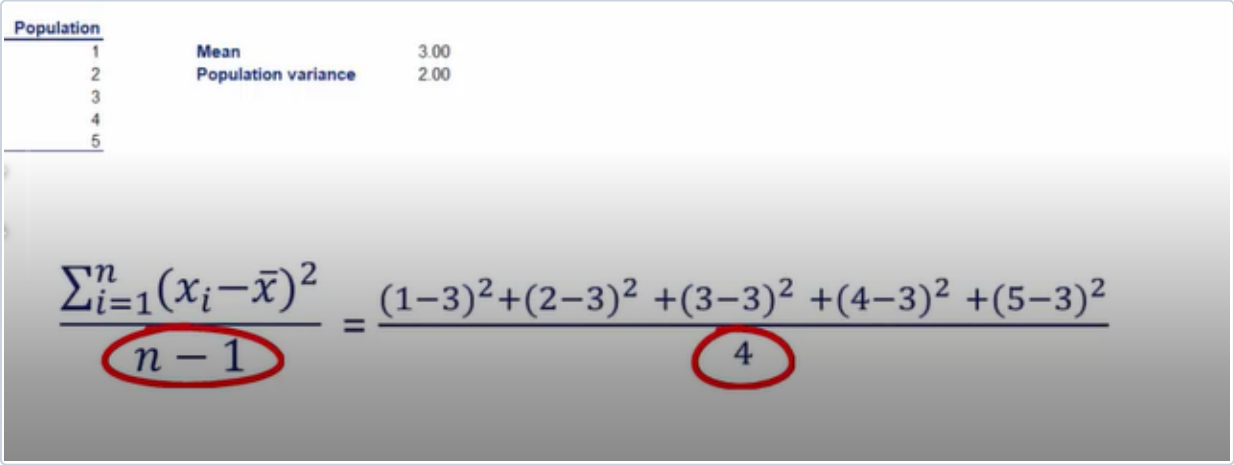
$$\sigma^2 = \frac{10}{5} = 2$$

SPREADSHEET: =VAR.P(B4:B8)

D	E
Mean	3
Population variance	2

NEXT › Step 4: the sample version uses n − 1

Step 4: the sample version uses n − 1



ORIGINAL SLIDE 11

TAKEAWAY

Dividing by 4 instead of 5 makes the sample variance **larger** than the population variance (2.5 vs 2). Use n − 1 whenever the data is a sample.

IN PLAIN WORDS

If the same numbers were a **sample**, the numerator stays 10 but the denominator becomes **n − 1 = 4**. The red circles on the slide mark exactly that change.

KEY IDEA

Sample variance s² = 10 ÷ 4 = 2.5

$$s^2 = \frac{10}{4} = 2.5$$

	Numerator	Denominator	Result
Population	10	N = 5	**2.00**
Sample	10	n − 1 = 4	**2.50**

SPREADSHEET: =VAR.S(B4:B8)

D	E
Sample variance	2.5

NEXT › The Excel workbook (page 13), then the cheat sheet

The Excel workbook: 02_Variance.xlsx

SHEET "VARIANCE"

B	D	E
1	Mean	=AVERAGE(B4:B8)
2	Population variance	=VAR.P(B4:B8)
3	Sample variance	=VAR.S(B4:B8)
4		
5		

CALCULATED VALUES

Quantity	Formula result
Mean	3
Population variance	2
Sample variance	2.5

Why the formulas matter. Editing any value in B4:B8 updates all three results at once, which is the quickest way to build intuition for variance. Try changing 5 to 10 and watch the variance jump.

EXTRA COLUMN "IMAGINARY POPULATION" (G4:G13)

G
1
1
1
2
3
4
5
5
5
5

WHAT IT IS

A second set of ten values the workbook lists next to the main data. **No formula in the file uses it**, and the lecture slides do not mention it.

For reference, calculated outside Excel: mean **3.2**, population variance **2.96**, sample variance **3.29**.

NOTE

Its purpose is not stated in the file, so I have not assumed one.

CHEAT SHEET

Variance on one page

THE FORMULAS

Population

$$\sigma^2 = \frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}$$

Sample

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

Use **N** when you hold the whole population and **n − 1** when you hold a sample.

THE SIX STEPS

1. Decide: population or sample?

2. Find the mean.

3. Subtract the mean from every value.

4. Square each difference.

5. Add the squares.

6. Divide by N (population) or n − 1 (sample).

Data 1, 2, 3, 4, 5	Result
Mean	3
Population variance σ²	2
Sample variance s²	2.5

KEY TAKEAWAYS

- Variance measures **how spread out** data is around its mean.
- A **sample** statistic only approximates the population parameter, and different samples give different values.
- Differences are **squared** so they are positive and cannot cancel out.
- Variance is in **squared units** (for example dollars²), which is hard to interpret. **Standard deviation**, the next topic, takes the square root to fix this.

IN EXCEL AND PYTHON

Tool	Population (÷ N)	Sample (÷ n − 1)
Excel	=VAR.P(range)	=VAR.S(range)
Python statistics	pvariance(data)	variance(data)
NumPy	np.var(data)	np.var(data, ddof=1)
pandas	Series.var(ddof=0)	Series.var()

COMMON MISTAKES

- Dividing by N when the data is only a **sample**.
- Forgetting that `np.var` defaults to ÷ N but pandas defaults to ÷ n − 1.
- Reading variance in the original units. It is in **squared** units.